HARISH S

Biomedical Engineering Student | IoMT & Embedded Systems | Data-Driven Healthcare Solutions

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PROFILE

Biomedical Engineering undergraduate specializing in medical device design, rehabilitation engineering, and wearable health tech. Experienced across biomedical instrumentation, embedded systems, IoMT, and cloud/AI-driven healthcare tools, with a published research paper, a second publication under review, and a filed patent application. Track record of award-winning projects and hands-on internships at Christian Medical College Vellore and CSIR. Eager to bring strong technical and problem-solving skills to an organization innovating in healthcare.

EDUCATION

Bachelor of Engineering (Honours) in Biomedical Engineering *2022 – 2026*
Dhanalakshmi Srinivasan College of Engineering and Technology | CGPA: 8.8

AISSE — Class XII, CBSE *2022*
376/500 | 75.2%

AISSCE — Class X, CBSE *2020*
435/500 | 87%

KEY PROJECTS

Neuro-Wave — 5G-Enabled Continuous EEG Monitoring System

- Led the winning team at the Naan Mudhalvan Statewide Hackathon 2025, securing 1st place (Feb 25, 2025).
- Designed an AI-integrated, wearable EEG system using 5G technology for real-time brain activity monitoring and early detection of neurological disorders.

Incubator Temperature Regulation System

- Built a PID-controlled system with IoT integration using an ESP32 microcontroller, LM35 sensor, and relays, with real-time data visualization via Adafruit IO.
- Published a research paper and filed a patent application based on this project.

Diabetic Foot Ulcer Detection Using Thermal Imaging

- Developed an AI-based model that detects and categorizes ulcer type from an uploaded image and recommends the corresponding treatment approach.

Breast Cancer Detection Using MATLAB

- Built an image-processing pipeline in MATLAB to preprocess and segment mammogram images, classifying tumors as benign or malignant with 96% accuracy.

Upper Limb Assist Device for Stroke Patients

- Designed an upper-arm rehabilitation device ("skateboard") in SolidWorks/AutoCAD equipped with an ArUco marker, in collaboration with Christian Medical College, Vellore.
- Programmed marker depth/position tracking in Python for real-time feedback, and used 3D printing to develop and validate the working prototype.
- Built a companion therapy game in Unity that uses live ArUco marker feedback to help stroke patients regain motor control and coordination.

Cloud-Based Smart Blood Bank System

- Designed a real-time blood bank platform on AWS/Azure with a live database for donor and recipient records, improving retrieval speed and transparency.

- Integrated IoT-based alerts to notify users of nearby donation camps and urgent blood requests, with a focus on data security and scalability.

Saviour Gate — IoT-Based ICU Hand Sanitization Compliance System

- Built an ESP32-based smart gate using dual ultrasonic sensors and a force sensor to detect entry and verify hand sanitization, with real-time LCD, LED, and audio feedback.
- Developed a companion Android app that authenticates users, displays live compliance parameters over WiFi, logs data to CSV in the background, and supports one-tap CSV export.

Stride Length Measurement System for Gait Analysis

- Designed a wearable dual-pod system (ESP32 + ultrasonic sensors + LiPo battery with charging module) worn on each ankle to measure inter-leg proximity and compute stride length.
- Implemented a pendulum-based biomechanical model using gyroscope angle integration and leg-length calibration to estimate stride length for both healthy individuals and rehabilitation patients.

PROFESSIONAL EXPERIENCE

Christian Medical College, Vellore — Intern *Jan 2026 – Present*

Council of Scientific and Industrial Research (CSIR), Taramani — Student Intern *Feb 2025 – Mar 2025*

- Built an Air Quality Monitoring System using an ESP32, MQ-135 gas sensor, I2C LCD, a 12V battery, and a buck converter to detect CO2, ammonia, benzene, and smoke.
- Programmed the ESP32 in Arduino IDE for sensor calibration and ADC-based signal processing, with real-time gas concentration alerts.

GirlScript Summer of Code — Open Source Contributor *Oct 2024*

Christian Medical College, Vellore — Rehabilitation Engineering Intern *Aug 2024 – Sep 2024*

- Studied stroke and spasm rehabilitation and used SolidWorks/AutoCAD to design the Upper Limb Assist Device and its companion Unity-based therapy game (see Key Projects).

Skill Vertex, Bengaluru — Cloud Computing Trainee *Jun 2023 – Sep 2023*

- Learned AWS and Azure cloud architecture fundamentals, deploying a static website to practice storage, serverless hosting, and cost optimization.
- Built the Cloud-Based Smart Blood Bank System as a major project (see Key Projects).

PUBLICATIONS

- AI-Based Breath Volatile Organic Compounds Pattern Recognition System for Early Health Risk Screening.
- Research paper on the IoT-based Incubator Temperature Regulation System (patent application filed).

ACHIEVEMENTS

- Young Scientist Award.
- Best Project Award — Detection of Breast Cancer Type Using MATLAB.
- Designed and deployed a food-ordering web application on AWS with a custom domain.
- 1st Place — National-level conference, Karpaga Vinayaga College of Engineering and Technology.
- 1st Place (internal round) — Smart India Hackathon; ranked in the All-India Top 100.
- 1st Place — Project Expo, 2nd year, Dhanalakshmi Srinivasan College of Engineering and Technology.
- 1st Place — Naan Mudhalvan Statewide Hackathon 2025 (Feb 25, 2025).

SKILLS

Programming: Python, Java, C, C++, JavaScript, HTML, CSS, SQL, Bootstrap, React

Tools: SolidWorks, AutoCAD, MATLAB, Arduino IDE, ESP32/Embedded Firmware, Android Studio (Kotlin), Unity, AWS, Microsoft Azure, Adafruit IoT, MS Office, Google Workspace

Specializations: Biomedical Instrumentation, Embedded Systems, IoMT, Data Science Basics, Diagnostic & Therapeutic Equipment, Medical Image Processing, Tissue Engineering, Forensic Science, Genetic Engineering

Soft Skills: Communication, Problem-Solving, Leadership, Multi-tasking

LANGUAGES

Tamil (Native) | English (Professional) | Hindi (Professional)